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SURVEY

# A Review of Human Synchronization Datasets

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**ABSTRACT** Human synchronization refers to the coordination in time of individuals' actions, movements, or physiological processes, often considered within a rhythmical context. This phenomenon can occur in various forms such as matching rhythmic movements, aligning speech patterns, or synchronizing biological rhythms. It plays a crucial role in social interactions, effective communication, sports, and musical performance. This work aims to reference accessible online datasets related to human synchronization, to increase their visibility and availability. Notably, it compiles 75 directly accessible datasets and 26 with indirect access to study synchronization in various contexts. It reviews all types of synchronization tasks and contexts found in the datasets. Results are clustered into four categories: discontinuous movement, continuous movement, social interactions, and musical performance. Then, this review explores the signals and acquisition methods used across human synchronization datasets, ranging from simple touchpads, and kinematic sensors, to expansive motion capture and hyperscanning technologies. This paper surveys objective measures of synchronization used in the gathered papers and reveals some heterogeneity in the quantification of synchronization. Furthermore, it points out a lack of annotated datasets, which limits the development of machine learning methods. By compiling accessible datasets, this review centralizes valuable experimental data, providing a foundation for more collaborative, comprehensive, and standardized approaches to studying human synchronization.

**INDEX TERMS** Sensorimotor synchronization, interpersonal coordination, human behavior, social signal processing, MIR, multimodal datasets, signal acquisition, time-series analysis.

## I. INTRODUCTION

### A. A MULTIFACETED PHENOMENON

Synchronization is a particular phenomenon that can manifest in very diverse systems ranging from clocks, flashing fireflies, planetary orbits, or circadian rhythms. This work focuses on the study of synchronization in human behavior and physiology. Some studies have examined the mechanisms that enable humans' remarkable ability to synchronize rhythmically with one another, whereas others have investigated the social benefits and outcomes of synchronized interactions. Multiple reviews have provided valuable insights

into specific aspects of synchronization, however, a unified perspective that bridges these dimensions remains limited.

Synchronization has been extensively studied in contexts ranging from simple rhythmic tasks (e.g., finger tapping) to complex activities, such as drumming and dancing. Repp and Su [102] reviewed how humans adapted their movements to external rhythms. It also describes the neural mechanisms underlying such behavior, involving sensory regions and the supplementary motor area, premotor, somatosensory, and motor cortices, but also cerebellum and basal ganglia. However, their research predominantly focuses on periodic coordination, leaving less regular and more complicated forms of synchrony underexplored. In addition to the study of its mechanisms, synchronization has been found to play a central role in facilitating social interactions.

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Physiological synchrony, as reviewed by Palumbo et al. [103], involves shared dynamics in heart rate, respiration, and skin conductance that enhance emotional attunement and social bonding in contexts such as caregiving and therapy. Similarly, behavioral synchrony is a fundamental basis of early development and cooperation. As reviewed by Leclère et al. [104], coordinated behaviors, such as gaze and vocal patterns in parent-child interactions, promote emotional and cognitive growth. Research also shows that synchronization directly benefits social bonding and cooperation. Research by Wiltermuth and Heath [105] further reveals that individuals engaged in synchronous activities, such as singing or marching together, reported higher levels of trust, group cohesion, and cooperation. Similarly, Kirschner and Tomasello [106] found that children participating in synchronous musical activities displayed enhanced cooperative behaviors compared to those in nonmusical contexts. At the neural level, Hasson et al. [107] described brain-to-brain coupling during communication, where shared neural activity enables mutual understanding and coordination, linking biological mechanisms to observable social behaviors. Overall, research in this field suggests that synchronization across behavioral, physiological, and neuronal levels promotes positive social interactions and outcomes [108]. In their review, Ayache et al. [109] focused on spontaneous interpersonal coordination, which refers to the uninstructed and natural alignment of human behaviors during social interactions. The authors emphasize a lack of a clear distinction between multiple synchrony definitions. They also review multiple heterogeneous paradigms used to study this phenomenon, including rhythmic activities, structured and unstructured conversations, and collaborative problem-solving tasks.

Furthermore, synchronization is a cornerstone of various individual and group activities, such as sports, dance, and music. In that scope, mastering different synchronization patterns such as in-phase and anti-phase synchrony and leader-follower relationships are common in art performing [110].

Synchronization manifests itself across multiple levels and many timescales. The terms synchrony, mimicry, coordination, and entrainment all try to catch and represent different aspects of human behavior. These variations in definition are closely linked to different methods and measures used in their respective studies. The following section explores the key definitions of these concepts as found in the literature.

## B. DEFINITIONS

### 1) ENTRAINMENT

In physics, synchronization refers to the behavior of coupled self-sustained oscillators, which are systems that can maintain periodic motion. These oscillators exhibit two key properties. First, as dissipative systems, they require a continuous energy input to sustain their oscillations. Second, they oscillate at their so-called natural frequency along a stable limit cycle. This cycle ensures that their amplitude is

robust to minor perturbations and that their phases remain free. When two self-sustained oscillators interact via a coupling mechanism, they can synchronize their phase and frequency under suitable conditions, a phenomenon known as “entrainment” [111]. Whether entrainment occurs depends on factors such as the strength of the coupling and the frequency detuning, which is the difference between the oscillators’ natural frequencies.

### 2) SENSORIMOTOR SYNCHRONIZATION (SMS)

The above physical theory has been applied with notable success in modeling human rhythmic movements. Indeed, some human movements exhibit the above-mentioned properties; for example, walking is a self-sustained, periodic, and stable oscillating motion. Sensorimotor synchronization (SMS), or action perception, describes the human ability to synchronize rhythmic motion with rhythmic stimuli. Common tasks involve tapping with a finger in time with an external rhythmic cue such as a metronome, but one can also pace the gait in the same way [112]. The sensory perception of these auditory, visual, or even tactile stimuli acts as a coupling mechanism, and if the conditions are met, the human motion is entrained to the stimulus frequency.

### 3) INTERPERSONAL COORDINATION (IC)

When people interact, they perceive and use social stimuli to produce and adapt their behavior to seamlessly interact. Interpersonal coordination refers to the interdependence of two or more people during social interactions, and can be subdivided into two main concepts: interpersonal synchrony (IS) and behavioral mimicry (BM) [113]. Although these two concepts differ at various points, they are not mutually exclusive and can overlap. Both shape the quality of social interactions [114].

3.i) Interpersonal synchrony (sometimes termed entrainment), as defined by Chetouani et al. [115] involves the dynamic and reciprocal adaptation of the temporal structure of behaviors between interactive partners. The authors insist that synchrony is dynamic in the sense that the important element is timing rather than the nature of the behaviors. Manifestation of synchrony occurs at different levels, including nonverbal behavior (e.g., facial expression, body movements, paralinguistic elements of speech), and covert ones like physiological hormonal and neural activities [116]. Both invisible biological rhythms and behavioral acts are subject to temporal alignment. This synchrony has been hypothesized to facilitate the processing of social signals [117].

3.ii) Behavioral mimicry on the other hand deals with the spontaneous imitation of gestures, postures, mannerisms, and other motor movements [118]. Its study usually consists of analyzing individuals in social interactions through their reciprocal mirroring of gestures on a 3 to 5-second temporal window. This includes small motor movements such as body posture and face touching.

### C. CHALLENGES, GAPS, AND OPPORTUNITIES

The work of Ėerekovi  [119], which focuses on gathering multimodal datasets of human behavior during social interactions, summarizes the key findings, challenges, and future directions in social signal processing. This constitutes an important step towards the automatic understanding of social interactions. This emphasizes the importance of capturing diverse forms of nonverbal behavior using robust multimodal datasets. By integrating behavioral, physiological, and neural data, such datasets offer a framework for analyzing synchronization in its full complexity. This underlines the need for interdisciplinary approaches and comprehensive annotations to deepen our understanding of how humans behave across different contexts and levels of interaction. However, this review focuses on social interactions, which only represent a specific context where humans can synchronize. Recognizing that adjacent research fields might get insight from each other and adopt a common unified and standardized framework. Therefore, we propose the current human synchronization dataset review, which to the best of our knowledge is the first to study synchronization with such a wide scope, including contexts such as social interactions, controlled SMS, sports, dance, and musical performance.

Building on these perspectives, our work aimed to summarize interdisciplinary insights and gather multimodal datasets of human interactions across diverse contexts. By extending the focus beyond social interactions to include domains such as Sport and Art, this review contributes to a broader understanding of synchronization and its role in human behavior.

Moreover, although numerous studies have collected relevant experimental data to analyze human synchronization, many are not readily available online. Some studies provided direct access to both raw and processed data; however, others restricted their access, requiring researchers to submit requests to the original authors. This practice slows the progress of research progress and hinders reproducibility. The lack of immediate access to datasets is also at odds with the FAIR (Findable, Accessible, Interoperable, and Reusable) principles, which advocate for the open sharing of research data to facilitate transparency and reproducibility. This effort aims to enhance reproducibility and facilitate further research in this field.

### D. CONTRIBUTION OF THIS WORK

The main objective of this review is to reference available online datasets related to human synchronization to enhance visibility. Reviews and collections of open access datasets have demonstrated valuable impact in other fields of research such as human-action recognition [120], brain-computer interfaces [121], or WiFi fingerprinting [122]. The scope of this work includes datasets from all types of human synchronization studies, ranging from simple and controllable setups of SMS within a rhythmic context to the more natural and complex ones of everyday life. Moreover,

we extend this scope with numerous audiovisual datasets of people dancing or playing instruments coming from the Music-Information-Retrieval (MIR) and Dance-Information-Retrieval fields (DIR). While these datasets were not initially collected for studying synchronization, they can be useful for testing measures and developing automatic methods. Datasets from fields that do not directly involve human interaction, such as the synchronization of complex networks involving computer systems, telecommunications, or real-time collaborative applications, were not included.

Meanwhile, we aimed to compile the different data acquisition methods, along with the various techniques to study them, as found in the literature. This provides insights into choosing appropriate synchronization measures, depending on the nature of the available data and the research question, that is, the specific aspect of synchronization being investigated.

All datasets and studies discussed in this review are compiled in a publicly available file on GitHub.<sup>1</sup> This document provides key details about each dataset, including the synchronization task and context, the number of participants, the type of stimuli, the types of signals collected, the recording devices used, the synchronization measures applied, and the research question. Researchers are welcome to contribute newly collected datasets to support the ongoing update of this resource.

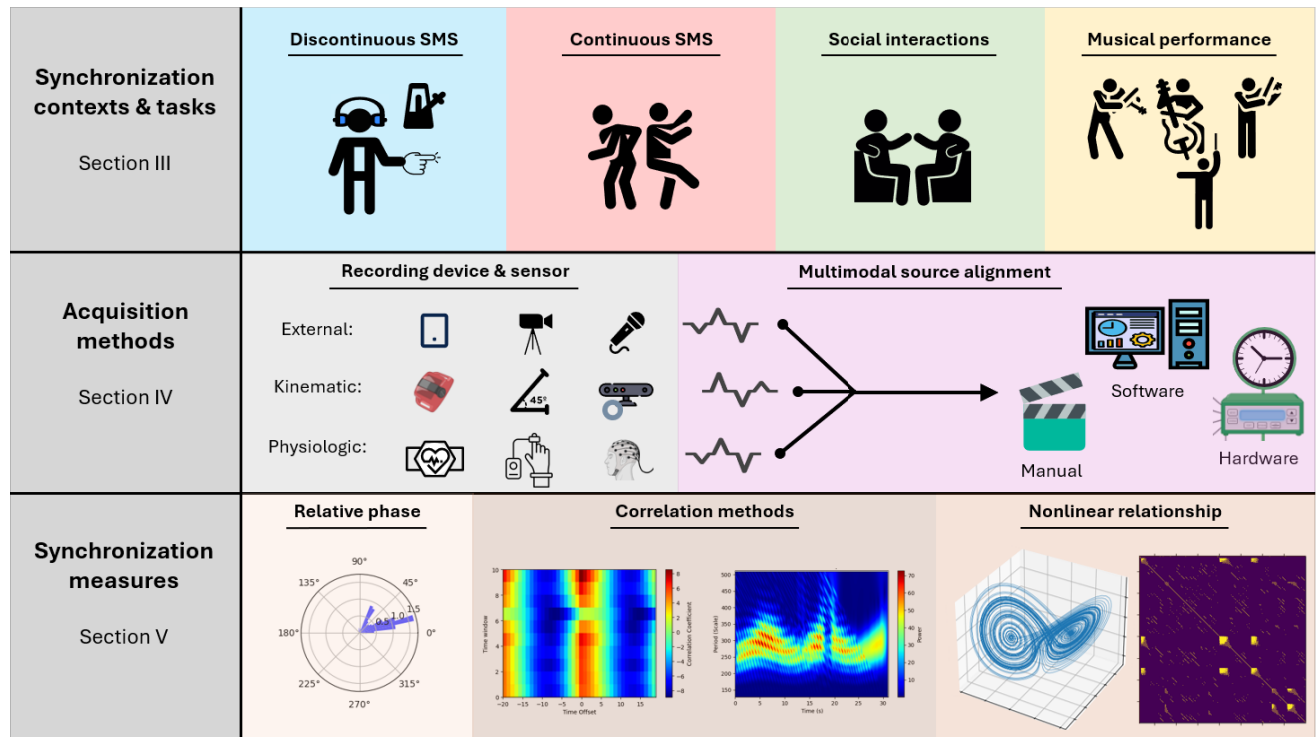
The paper is organized as follows: the dataset collection methodology is presented in Section II. Sections III, IV, and V describe the results of this study, as follows. Section III reviews the different synchronization tasks and contexts studied in the literature. Section IV describes the data acquisition techniques used to record the different aspects of human behavior. In Section V, we enumerate the multiple measures of synchronization used in each context. These different results are depicted in Fig. 1. Discussion regarding annotations and limitations of the gathered datasets is provided in section VI. Section VII concludes this work and envisions future works.

## II. METHODOLOGY

This review aims to compile a comprehensive collection of datasets related to human synchronization, guided by several key principles. Studies were selected based on their relevance to human synchronization in various contexts, including sensorimotor, musical, and interpersonal synchronization. In line with the FAIR (Findable, Accessible, Interoperable, and Reusable) principles, datasets were included if they were openly accessible or could be reasonably obtained through the author's request.

Furthermore, this review incorporates datasets from research fields beyond synchrony, such as DIR, MIR, and motion recognition, as they have the potential to contribute valuable insights to synchronization research. By broadening

<sup>1</sup>[https://github.com/TheoPulse/Human\\_synchronization\\_datasets.git](https://github.com/TheoPulse/Human_synchronization_datasets.git)



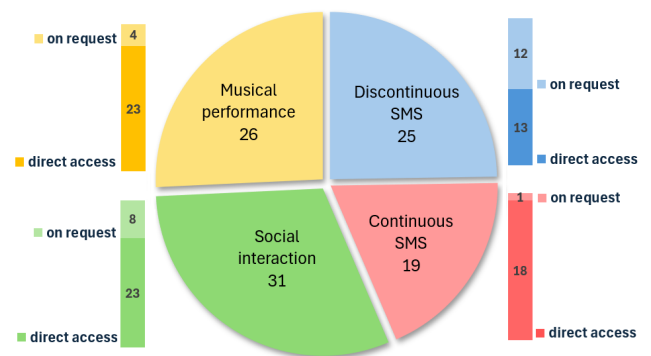
**FIGURE 1.** Graphical abstract of the main results, categorized in terms of contexts and tasks, acquisition methods and synchronization measures.

the scope, this review aims to ensure a diverse and inclusive dataset collection.

The dataset collection process began with a literature search using Google Scholar<sup>2</sup> and Google Datasets.<sup>3</sup> Keywords “sensorimotor synchronization,” “auditory-motor synchronization,” “beat synchronization,” “interpersonal synchronization,” “interpersonal entrainment,” “musical synchronization,” and “ensemble synchronization” were used in combination with the term “dataset”. Given the substantial number of results generated, we limited our search to the first 30 most relevant articles per keyword.

To extend this collection, the “similar articles” feature available on major scientific journal platforms, Springer, Nature, and Elsevier, has been employed. This step identified additional articles that did not appear in the initial search but were nonetheless highly relevant.

During the review, articles containing datasets from adjacent fields such as human movement and musical performance were systematically incorporated. This inclusion ensures a broader perspective of synchronization-related phenomena. Studies in which datasets were not openly accessible, but were available upon reasonable request, were also added to the collection. This systematic online search approach enabled the collection of a range of datasets spanning various aspects of synchronization research.



**FIGURE 2.** Datasets accessibility status and volume distribution among the 4 categories.

### III. CONTEXTS AND TASKS IN HUMAN SYNCHRONIZATION RESEARCH

A total of 101 datasets on human synchronization were collected. Table 1 lists the 45 raw datasets specifically collected to enhance further research, while Table 2 lists the 56 experimental studies that are based on the original data and described in a scientific paper. Among raw datasets and studies, 75 datasets provide direct online access. In others, authors explicitly specified their dataset is available upon reasonable request.

To better understand the diversity of synchronization behaviors, we propose classifying these datasets according to the nature of the recorded task. These tasks can differ

<sup>2</sup><https://scholar.google.com/>

<sup>3</sup><https://datasetsearch.research.google.com/>

**TABLE 1.** The different raw datasets of human synchronization identified in the review. Raw dataset's explicit task and context, and their accessibility status.

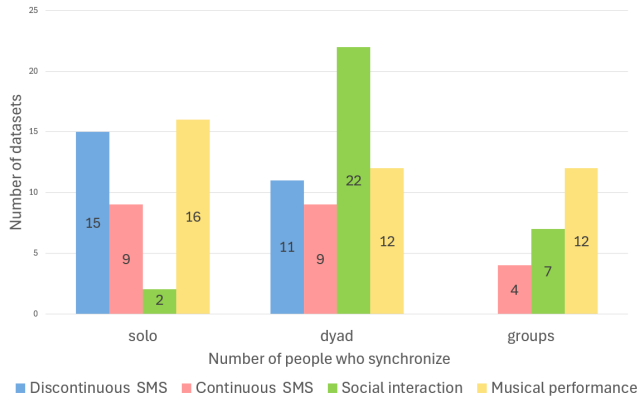
| Dataset Name                                            | Task and Context                               | Direct access |
|---------------------------------------------------------|------------------------------------------------|---------------|
| Music EEG Dataset - Tempo (NMED-T) [1]                  | Tapping                                        |               |
| AIST DANCE VIDEO DATABASE [2]                           | Dance                                          | x             |
| CMU Mocap dataset [3]                                   | Everyday life tasks                            | x             |
| PONE Walker synchronization [4]                         | Walking                                        | x             |
| MADS: Martial Arts, Dancing, and Sports Dataset [4]     | Martial arts / sports / dancing                | x             |
| UCF101: Dataset of 101 Human Actions [5]                | Human interaction, Playing Music, Sports       | x             |
| AlignNet: Dance50 dataset [6]                           | Dance                                          | x             |
| SEWA DB: A Rich DB for Emotion Research [7]             | Natural dyadic conversation                    | x             |
| Cardiff Conversation Database (CCDb) [8]                | Natural dyadic conversation                    | x             |
| 4D Cardiff Conversation Database (4D CCDb) [9]          | Natural dyadic conversation                    | x             |
| The MAHNOB Mimicry Database [10]                        | Dyadic conversation on politic                 |               |
| Conflict Escalation Resolution (CONFER) [11]            | Dyadic conversation on politic                 | x             |
| RECOLA: Multimodal Corpus of Interactions [12]          | Dyadic conversation during a video conference  |               |
| GFT Facial Expression Database [13]                     | Natural dyadic conversation                    | x             |
| K-EMOCON: multimodal DB for emotion recognition [14]    | Natural dyadic conversation                    | x             |
| IEMOCAP: interactive emotional dyadic Mocap DB [15]     | Natural dyadic conversation                    | x             |
| The MSP-IMPROV Database [16]                            | Conversational dyadic improvisations           | x             |
| MHi-Mimicry: DB for Mimicry Analysis [17]               | Dyadic conversation on a political topic       |               |
| AMI Meeting Corpus [18]                                 | Meeting conversation                           | x             |
| Danish NOMCO : multimodal interaction [19]              | Natural verbal interaction                     |               |
| RAMAS: Russian Multimodal DB of Dyadic Interaction [20] | Scripted scenarios                             |               |
| USC Creative DB of dyadic interactions [21]             | Theatrical improvisation                       | x             |
| MUMBAI: multi-person, board game interaction [22]       | Playing board games                            | x             |
| The NoXi database: novice-expert interactions [23]      | Screen mediated conversation                   | x             |
| The Tower Game Dataset [24]                             | Playing a game building a tower                |               |
| M-MS: DB of Dyadic Interaction in ASD [25]              | Children following music therapy activity      |               |
| AMIGOS Dataset [26]                                     | Watching videos                                | x             |
| the SEMAINE corpus [27]                                 | One way conversation responding to sal agent   | x             |
| ELEA corpus [28]                                        | Conversation in imagined survival scenario     | x             |
| Multi-Modal Music Performance URMP Dataset [29]         | Mus. perf. multiple instruments                | x             |
| Solos: A Dataset for Audio-Visual Music Analysis [30]   | Mus. perf. 13 instruments                      | x             |
| MOSA: Music Motion with Semantic Annotation DB [31]     | Mus. perf. piano or violin                     | x             |
| Expressive Musical gestures [32]                        | Mus. perf. piano or violin                     | x             |
| MUSIC dataset [33]                                      | Mus. perf. multiple instruments (11)           | x             |
| MUSIC-Extra-Solo [34]                                   | Mus. perf. multiple instruments (9)            | x             |
| C4S: Clarinetists for science [35]                      | Mus. perf. clarinet                            | x             |
| Enst-drums: audio-visual drumming DB [36]               | Mus. perf. drum                                | x             |
| EEP: Ensemble Expressive Performance [37]               | Mus. perf. string quartet                      | x             |
| Multimodal Guitar dataset [38]                          | Mus. perf. guitar                              |               |
| IEMP Cuban Son and Salsa [39]                           | Mus. perf. multiple instruments + voices       | x             |
| IEMP Malian Jembe [40]                                  | Mus. perf. jembe drumming                      | x             |
| IEMP North Indian Raga [41]                             | Mus. perf. voices + multiple instruments       | x             |
| IEMP Tunisian Stambeli [42]                             | Mus. perf. voices + multiple instruments       | x             |
| IEMP Uruguayan Candombe [43]                            | Mus. perf. drum ensemble performance           | x             |
| IEMP European String Quartet [44]                       | Mus. perf. String quartet (violin viola cello) | x             |

**TABLE 2.** The four categories of human synchronization studies identified in the reviews. Explicit task and context used in the studies providing the datasets and their accessibility status.

| Human synchronization category | Explicit tasks                                            | Direct access                                        | Access on request                                              |
|--------------------------------|-----------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------|
| Discontinuous SMS              | Tapping                                                   | [45], [46] [47] [48] [49] [50] [51]<br>[52] [53]     | [54] [55] [56] [57] [58] [59] [60]<br>[61] [62] [63] [64] [65] |
|                                | Clapping, whispering                                      | [66] [67] [68]                                       |                                                                |
| Continuous SMS                 | dance, bouncing                                           | [69] [70] [66]                                       |                                                                |
|                                | Hand balancing, finger oscillation                        | [71] [72] [73] [74] [75]                             | [76]                                                           |
|                                | Pendulums                                                 | [77] [78]                                            |                                                                |
|                                | Walking, running, other                                   | [79] [80]                                            |                                                                |
| Social interaction             | Conversation, watching videos, interaction, playing games | [81] [82] [83] [84] [85] [86] [87]<br>[88] [89] [90] | [91]                                                           |
| Musical performance            | Playing instruments                                       | [92] [93] [94] [95] [96] [97] [98]                   | [99] [100] [101]                                               |
| Total                          |                                                           | 39                                                   | 17                                                             |

at different levels, including the type of movement studied and its context (i.e., whether synchrony is assessed in naturalistic dyadic interactions, in more controlled settings, or in musical performance). Through this synchronization

task lens, we categorized these datasets into four categories - roughly equivalent volume-wise (see Fig. 2) - which we will refer to as: discontinuous SMS, continuous SMS, social interactions, and musical performance. Each of the four



**FIGURE 3.** Number of datasets studying, one, two or multiple synchronizers, in each category.

categories reflects a unique facet of how individuals or groups synchronize their actions, either solo, in dyads (groups of two), or larger groups (see Fig. 3), spanning a wide variety of tasks and contexts and synchronization behaviors.

Datasets might include rhythmic motor synchronization tasks such as tapping to discrete stimuli, as well as more continuous and flowing actions such as walking and dancing. They also include more complex forms of social and emotional coordination, such as those observed in conversations or musical group performances. Below, we describe synchronization tasks and types and contexts that can be found in each of these four categories.

#### A. DISCONTINUOUS SMS

In this study, 25% of the datasets (see Fig. 2) contain records of individuals performing a rhythmic movement along with the presence of rhythmic auditory stimuli, i.e., an SMS task. The name of this category, discontinuous SMS, refers to the nature of the stimuli. The most common choice for these stimuli was the auditory metronome. Participants were invited to align their motions on the discrete landmarks produced by the metronome. Most of the time, SMS is analyzed through finger tapping, however, it is not restricted to it. We also find other behaviors being studied, such as clapping [66] or whispering [56]. SMS tasks are practical due to their simplicity and controllable settings. In such setups, one can manipulate the complexity of the rhythm, ranging from simple metronomes to more syncopated ones, or even music. Also, the rate of the metronome can be adjusted, and the participants are invited to synchronize at different frequencies. Sometimes the frequency of the metronome varies over time, gradually slowing down or speeding up; this is referred to as a drifting metronome [47]. It is common for experimenters to use auditory stimuli for their SMS task, but some studies have focused on visual or tactile stimuli [58], as illustrated in Fig. 4. The tasks in this category were designed to assess different aspects of human behavior. Ranging from self-paced activities to

complex synchronization challenges. Below, we review the main ones:

i) Self-paced are activities in which participants perform actions at their rate, without being externally timed or synchronized with a stimulus or another person [46]. The pace of the task is determined by the participant's internal timing mechanisms.

ii) Synchronization activities involve participants performing actions in time with an external stimulus [68] or with each other [62], or both [47]. These tasks are designed to measure how well individuals can align their actions with the timing of an external cue and the modeling of human synchronization.

iii) Sync-continuation paradigms require participants to synchronize their actions with an external stimulus for a period and then continue performing the motion at the same pace after the stimulus has stopped [53]. This type of task assesses the participant's internal timekeeping and his ability to maintain a rhythm independently, from memory.

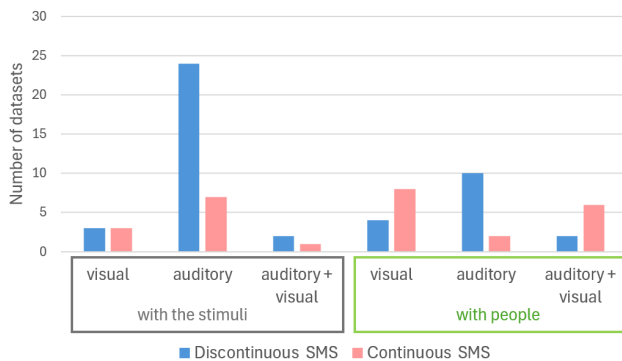
iv) The perturbation paradigm involves introducing unexpected changes or disruptions to the environment or task conditions to observe how participants adjust their actions to maintain synchronization. These tasks are used to study dynamical stability, adaptability, and the mechanisms underlying synchronization.

v) Beat-finding tasks require participants to listen to music and tap along with their fingers on the beat of the music. These tasks are used in beat-perception studies, and also in MIR for developing algorithms that can detect musical beats [52].

Finally, these tasks mostly differ in their choice of stimuli which can be a metronome, music, or another person's movements with the possibility of perturbation, acceleration, or even stopping.

#### B. CONTINUOUS SMS

The continuous SMS category accounted for 19% of the datasets (see Fig. 2). This category includes continuous rhythmic movements, such as walking together [70], oscillating pendulums [77], or dancing to music [70]. We also find experiments in which individuals perform non-rhythmic movements, such as raising their arms [75], playing mirror games with their hands [74], or performing Tai-chi routines [80]. In contrast to the first category, discrete landmarks are not easily definable in these activities. Most of these datasets (14/19) allow the study of interpersonal coordination. Among them, the modality of the coupling between the participants is either visual, auditory, or both, as illustrated in Fig. 4. This figure distinguishes coupling modality depending on the nature of the rhythmic external context. This rhythmical context can either be a stimulus, e.g., music, or another individual performing a rhythmic motion. In the former case the coupling is unidirectional, while in the latter, one can observe mutual adaptation among coupled individuals. Some studies analyze the impact of coupling modality and direction on the synchronization dynamics outcome [72], [76]. In [77], interpersonal coordination is studied



**FIGURE 4.** Number of datasets, sorted by sensory conveying the stimulus, with an external periodic event or between individuals, for discontinuous and continuous SMS categories.

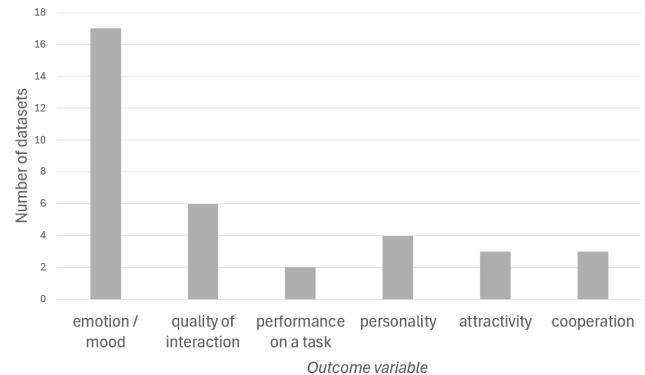
within groups of 7 participants instead of dyads. Variation of the coupling network topology impacts synchronization dynamics.

### C. SOCIAL INTERACTIONS

The social interaction category is the largest with 31 studies and datasets (30%, see Fig. 5). It compiles datasets for studying IC between two or more individuals during social interactions. Half of the datasets were obtained from the affective computing field. Affective computing aims to enable computers and systems to recognize, interpret, simulate, and respond to human emotions in a way that enhances interaction between humans and machines. Datasets from this field can be great resources for analyzing synchrony in social interactions at the emotional level. The social interactions datasets gathered here allow us to study IC in various contexts such as free and scripted dyadic conversations (22/31) [14], speed-dating (3/31) [85], playing board games [91] (2/31). These datasets allow us to study various aspects of IC such as behavioral mimicry and interpersonal synchrony at emotional and physiological levels. In comparison to the above two SMS categories, the research questions here are generally different. Instead of understanding the mechanism that affects the strength and the stability of the synchronization, the goal here is to study and compare how these behavioral alignments are linked to an outcome variable such as cooperation [22], personality [91], attractiveness [85], quality of interaction, and emotion (see Fig. 5). These variables are either obtained through self-reported questionnaires, or semi-automatically and manually annotated by experts. Datasets from this category provide many annotations from multiple experts, where consistency, accuracy, and inter-rater reliability are controlled.

### D. MUSICAL PERFORMANCE

Datasets from the musical performance category represented 26% of this review and spanned two adjacent interdisciplinary fields. The first half focuses on music interpersonal entrainment, which studies how individuals synchronize their



**FIGURE 5.** Volume of social synchrony datasets containing annotation of different social outcomes.

movements, emotions, and behaviors in response to music. Indeed, playing an instrument in groups emphasizes both social and sensorimotor synchronization skills. Musicians must coordinate and integrate fine behavioral cues to adjust their music and create a musical piece together. The second half derives from the Music Information Retrieval (MIR) and Dance Information Retrieval (DIR) domains, which use computational techniques to analyze and extract patterns from music and dance. Datasets from this category typically include diverse musical genres such as jazz [97], classical [29], Indian raga [41], Malian jembe [40], etc. The performances feature a wide range of instruments, including drums, violins, cellos, clarinets, pianos, guitars, and vocals. They may involve solo performances [30], duos [95], or groups ranging from three to sixteen people [92]. The datasets vary in richness, including single audio/video files, multitrack audio [29], annotated musical scores, and motion capture data [32], offering a holistic view of music-driven interactions.

## IV. DATA ACQUISITION METHODS

This section reviews the different data acquisition methods and technologies used to study human synchronization in reviewed studies. Capturing the intricate timing and dynamics of movements and interactions is technically challenging, particularly in a multimodal context. In Fig. 6, we observe the distribution of the acquired data among the four synchronization categories described above. We found eight types of data, namely: physiological (heart rates, skin conductance, etc.), event time-series, EEG (i.e., brain electrical activity), kinematic (i.e., obtained from sensors like accelerometer and potentiometers), audios, simple videos, pose estimation (i.e., skeletal segments position), and motion capture which records position from light markers put on individuals' key articulation. Some datasets combine multiple types, resulting in overlap among categories in Fig. 6. One can observe that social interactions are often captured using simple video, while motion capture is predominantly used for complex continuous movements or musical performances. Discontinuous

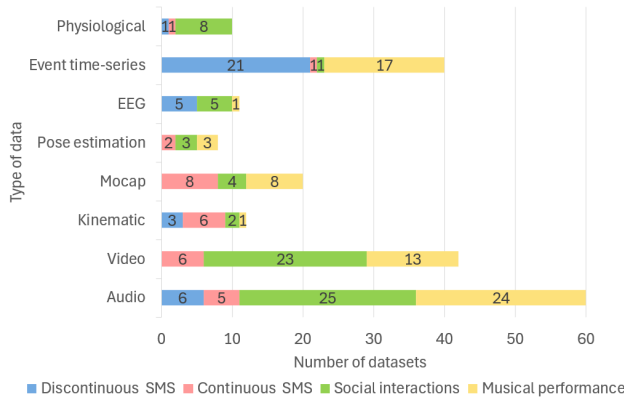


FIGURE 6. Type of acquired data among the 4 synchronization categories.

SMS studies frequently rely on event time-series data for their analysis.

### A. ACQUISITION AT VARIOUS LEVELS

Interpersonal coordination manifests at multiple levels. Numerous key signals help to understand how individuals coordinate and interact in social, collaborative, or task-oriented settings at different levels. These signals can be categorized into three large categories: behavioral, physiological, and neural [116].

#### 1) BEHAVIORAL LEVEL

##### 1.i) Recording simple & deliberate movements

At the behavioral level, there are datasets of rhythmic synchronization of either continuous or discontinuous movements. As shown in Table 3, most discontinuous SMS datasets were recorded using event-recording devices. Some are collected using force-sensitive devices such as pads and keyboards. These devices record MIDI information, including force and velocity. Two studies recorded the movements using accelerometers. Less often, finger taps were recorded using a microphone placed near the tapping zone. This method requires additional processing of the sound envelope to extract the movement onsets. It also allows experimenters to record speech during SMS tasks [56]. In [60], the authors used the Rhythm ExPeriment Platform (REPP) which is an online-based technology that can measure the asynchrony of participant's taps along music clips. It has the advantage of using the built-in microphone of the participant's computer, allowing the experimenter to collect more data easily. Five studies record the tapping times using computer keyboards. The "other" category contains devices such as tactile screens, electro-optic sensors, and handmade force-sensitive devices that record pressure but do not convert it to MIDI.

##### 1.ii) Recording richer & deliberate movements

For the study of more complex and continuous movements, when the precise timing of the SMS task is not the only feature of interest, experimenters record kinematic information. Potentiometers can be used to record the angular

TABLE 3. Devices used for recording tapping motion in discontinuous SMS tasks.

| Collection device | Studies                                        |
|-------------------|------------------------------------------------|
| Pads & keyboards  | [45], [54], [47], [48], [55], [51], [61], [65] |
| Accelerometers    | [64], [66]                                     |
| Microphone        | [49], [56], [67], [68]                         |
| Computer keyboard | [52], [53], [58], [60], [62]                   |
| Other             | [1], [46], [57], [59]                          |

position of limbs. Accelerometers are common practical devices that allow the study of movement trajectories [79]. These devices allow the recording of the position, speed, and acceleration of the point they are attached to.

In some studies, the movement of an object is required, thus potentiometers are attached to a pendulum to record its kinematics over time [77]. Another approach involves motion capture (mocap) [69]. Multiple reflective markers are placed on the participant's body points of interest, and sensitive cameras track those markers to digitally record movements in space. This technology is expensive and requires the experiment to be performed in a laboratory.

The use of simple cameras to record whole-body movements is common, as they allow recording of whole-body movements in more natural settings. However, additional processing is required to extract the coordinates of the points of interest. Some datasets include pose-estimation data. Pose estimation refers to the process of identifying the position and orientation of a human body (or other object) in an image or video frame. It typically involves detecting key points on a body (e.g., joints such as elbows, knees, and shoulders) and constructing a skeleton-like representation of the subject. Some datasets provide videos of movements along with their 2D [6] (height and width) or 3D (+depth) pose estimation [94].

##### 1.iii) Recording fine behavioral cues

Moving beyond deliberate actions, unconscious subtle behavioral adjustments also play a significant role in interpersonal coordination, especially in social interactions. Videos are also a common type of data for studying synchrony during social interactions. Indeed, the recording of multiple people's bodies and faces while they are having a conversation along with the use of automatic video analysis can help identify subtle simultaneous behavioral adaptations, such as gestures, smiles, gazes, etc. [10]. Paralinguistic speech features are also a nonverbal dimension where interacting individuals may synchronize. This is referred to as vocal entrainment and consists of the adaptation of paralinguistic features of speech such as pitch, rhythm, tone, and speech rate [91]. Simple microphones or cameras, along with post-processing, allow the acquisition of meaningful signals.

#### 2) PHYSIOLOGICAL LEVEL

Looking at the physiological level heart rate and electrodermal activity are also commonly studied in interpersonal synchrony studies. These autonomous physiological signals may spontaneously synchronize with others when they inter-

act. They can reflect emotional arousal, allowing us to study how individuals synchronize their states at the emotional level [103]. Data are collected using electrocardiogram (ECG), electromyogram (EMG), and electrodermal activity (EDA) sensors placed on the skin of individuals.

### 3) NEURAL LEVEL

The last level of synchronization reviewed in this work is neural activity. It is indeed the subject of a specific field of research called inter-neural synchronization. Common techniques for measuring brain electrical activity include EEG, fMRI, and fNIRS. A particular collection of datasets was obtained from the hyperscanning studies. Hyperscanning is a technique that involves the simultaneous recording of neural activity from two or more individuals as they engage in a shared task or interaction [123]. Half of these datasets (4/10) are recorded while individuals engaged in joint motor tasks [62], or during dyadic social interactions (5/10) [82], and the last one through joint piano playing [101]. This approach allows researchers to explore inter-brain connectivity and interpersonal synchrony, providing insights into how brains align during communication, cooperation, and joint action.

## B. MULTIPLE SOURCES ALIGNMENT STRATEGIES

In the context of synchronization studies, experimenters are interested in capturing multiple modalities and perspectives. To achieve this, they may use separate devices with different sampling frequencies. This commonly leads to the technical issue of synchronizing multimodal data. To do so, at least two strategies exist. Hardware synchronization of acquisition devices and post-production data streams re-synchronization. Each comes with its advantages depending on the type of data, recording setup, and desired precision.

i) Hardware-based methods ensure precise alignment by embedding timestamp information or using external time references. This ensures that both audio and motion capture systems are synchronized at the hardware level by sharing the same time reference across different computers. In the multimodal guitar dataset [38], a world clock generator system was used to control the sampling instants of the capture devices. In EEP - ensemble expressive performance [37] - a linear timecode (LTC) system is used to synchronize 100Hz Mocap data with 44KHz audio data.

ii) Manual: In setups in which data is recorded across multiple devices, such as motion capture, audio, and video, an event-based method such as a clapperboard might be employed. For instance, in a study on the knee flexions of sax players [99], a clapperboard provided a clear audiovisual signal at the start and end of each take. The visual cue in the video and the sound captured in the audio allowed precise synchronization in post-processing by aligning these signals across the different data streams.

iii) Software-based: This relies on extracting features or shared events from different data streams and aligning them

algorithmically. For instance, in the MOSA dataset [31], chroma sequences were first extracted from each audio recording. Then, temporal alignment was achieved by applying cross-correlation between the different data streams. The lag value at which the correlation is maximal is considered the time difference between the two audio recordings.

Increasing the number of modalities with high-frequency signals such as EEG, motion capture, and video data inevitably complicates the process of aligning monitored signals altogether. This raises the concern of developing a scalable solution for acquiring large multimodal datasets. The lab streaming layer (LSL) [124] is a software-based method that allows precisely recording synchronized multimodal human neurobehavioral data streams. LSL separates data collection from data storage and processing. This decoupling allows the system to continuously stream data in real-time without being constrained by storage or processing bottlenecks. Moreover, it is compatible with external scalable storage tools.

## V. MEASURING SYNCHRONIZATION

This section provides an overview of the most common measures and algorithms identified in the gathered studies for quantifying synchronization. Notably, multiple measures exist to assess synchronization between two or more signals, each computing distinct scores that capture different but related aspects of coordination. This diversity complicates the selection of the most appropriate measure. Comparative analyses on real and synthetic data [80], [125], [126], [127] have shown that no single measure universally outperforms the others; each has its strengths and limitations. We review the most used measures below, considering the properties of the signals and the specific aspect of coordination researchers aim to explore.

### A. BETWEEN PERIODIC MOTION AND A REFERENCE SIGNAL

In numerous studies of this review, mostly coming from the discontinuous SMS category, participants are asked to align a periodic motion along with the onsets of a metronome. Their motion produces an event time series that is compared with a reference time series, such as that produced by a metronome. Deviations from it are termed asynchronies. Simple statistics of this series of asynchronies such as mean and standard deviation, give an overall degree of synchronization. Their ratio, which is called consistency, represents how isochronous the tapping was within a given trial [46]. The series of inter-tap intervals (ITI) can also be analyzed [67]. Autocorrelation methods may reveal serial dependencies and distinct timing strategies [58]. Time series of periodic signals, such as series of asynchronies or ITIs - can be projected on a unit circle. This produces a series of relative phases requiring circular statistics, and naturally fits in with the oscillator modeling approach. The circular mean and variance allow us to gain insight into the quality of synchronization. The mean resultant length, or circular

mean, is also called the phase locking value (PLV). It outputs a single scalar from two times series and is considered as the synchronization index [56].

When analyzing the synchronization of a continuous signal with a reference, one could extract discrete events from peaks of amplitude and use the PLV measure mentioned above [67]. Another technique consists of extracting the phases of the signal using spectral methods such as Fourier, wavelet, or Hilbert transform [72]: Once the phases are extracted, the PLV can be computed.

### B. DEALING WITH MULTIPLE AND COMPLEX SIGNALS

When analyzing multiple and richer signals, such as physiologic and unconstrained nonverbal behavior signals in naturalistic conversation, correlation methods [85], either in the temporal domain (cross-correlation) or in the spectral domain (coherence), can help reveal the codependency of signals. These techniques are of interest owing to their simplicity of implementation. These are global measures that output a single scalar representing the similarity of two time series. Both methods assume stationarity and linear dependence of the signals. Cross-correlation can produce a heatmap of correlation values between two time series at various time-lags. This method is useful for studying the leader-follower strategy [57]. Cross-spectral coherence on the other hand computes frequency overlap between the signals. It can be of interest when the signals are rich, i.e., composed of multiple frequencies, such as the brain electrical activity [89], or when one is interested in studying the phase relationship at different frequencies [72].

These two methods can be adapted to weaken their assumptions regarding time-series properties. By studying signals over temporal windows, the global stationarity assumption can be reduced to local stationarity over a given window size. Thus, this method is versatile. In [92], the windowed fast Fourier Transform was used to identify the different notes and timings produced by violin players. Another approach consists of using the cross-wavelets transform (CWT), using for instance the Morlet wavelets. CWT results in direct time-frequency decomposition, allowing for the examination of interactions at multiple timescales. One can find an application of this method combined with the cross-coherence used to quantify the synchronization of heart rate activities in [62].

### C. REVEALING STRUCTURE AND DIRECTIONALITY IN SIGNALS

When two or more people interact, their behavior will likely be non-strictly periodic and stationary. For instance, in joint synchronization-continuation tasks, the participants synchronize their motion along with a periodic reference; then the reference is taken out and the participants must continue to produce their rhythmic motion trying to keep the same pace. A practical tool for this use case is recurrence quantification analysis (RQA). RQA is a nonlinear graphical and quantitative method that reveals patterns of recurrence

or repeated states in a system, making it particularly useful for exploring synchronization, causality, and structure in dynamic interactions. Quantitative measures, or dynamical invariants, include recurrence rate which corresponds to the proportion of recurrent points in the plot [48], [55]. This can also be of great use in studying more complex, nonlinear, and non-stationary time series such as the one produced by head movements during dyadic conversation [83].

Granger causality analysis (GCA) also goes beyond correlation by assessing directional influences. It tests if past information from one time series can help in improving the prediction of the future of another time series. This has been applied to determine whether activity in one brain region or individual better predicts activity in another than vice-versa [82], or to understand the directionality of communication in professional quartets of violin players [93].

For the detection of directionality between more than two signals, a net causation entropy measure has been proposed in [78]. It is based on Shannon entropy and quantifies the net information flow between one player and all the other players and is taken as the influence leadership index in this study.

## VI. DISCUSSION

### A. SYNCHRONIZATION: A COMPLEX PHENOMENON

#### 1) TOWARDS A MULTILABEL VIEW ACROSS TASKS, ACQUISITION, AND MEASUREMENT

The results section reveals the diversity of contexts found in the literature that study the synchronization phenomenon, seen from the dataset viewpoint. We classify our results according to three main perspectives: tasks, acquisition, and measures. The classification by task reveals four main balanced categories: discontinuous SMS, continuous SMS, social interactions, and music performances. The acquisition section shows a variety of tools that focus on detecting different realizations of synchronization, on behavioral, physiological, and neural levels. Devices vary widely, ranging from simple touchpads, microphones, and cameras to wearable kinematic and physiological sensors, to expensive mocap or hyperscanning technologies. These tools typically output time-series data. The last section reviewed common measures used for analyzing such time series, depending on their complexity.

Sections III, IV, and V reveal a great deal of heterogeneity across the different splitting, namely tasks, acquisition, and measures. For example, a tapping task acquisition ranges from only time onsets with computer keyboards [52], to MIDI information with musical pads [51], to full joint angle, speed, and acceleration information with accelerometers [64]. Interpersonal coordination is sometimes studied through discrete SMS tasks [54], continuous SMS tasks [71], or in less rhythmic settings such as natural conversation [83]. Regarding measure, for the same signal properties, multiple approaches are used in the literature. All of them present distinct pros and cons and reveal potentially different aspects of synchronization. For instance, leader-follower behavior is

sometimes assessed through cross-correlation [57], Granger causality [94], or even RQA [45].

Choosing the appropriate synchrony measure remains an open question. The effectiveness of a given measure depends on several factors, including the properties of the signals, the presence and type of noise, and the underlying system generating the data. Using synthetic data is a practical way to compare different measures and analyze how changes in signal properties affect them. A comparative analysis on synthetic data by [126] shows that each metric performs differently depending on the scenario. Notably, the authors stress that no single measure consistently outperforms all others, and they might complement one another.

However, synthetic data often simplifies real-world complexity and may not fully capture the characteristics of natural signals. In real-world scenarios of human synchrony research, the diversity of contexts and signal types adds another layer of complexity, making it difficult to establish a unified framework across domains.

The multiSyncPy package [125], which allows for the comparison of six multivariate synchrony measures on both synthetic and real data, highlights this issue: measures that perform well in controlled conditions may struggle in real-world scenarios. A promising direction for future work is to generate more realistic synthetic datasets that incorporate complex properties such as nonlinearity and quasi-periodicity to better evaluate synchrony measures in applied settings.

Considering these challenges, the proposed dataset collection simplifies access to a diverse collection of datasets covering different contexts and signal types. This should help evaluate synchrony measures more systematically and improve our understanding of their strengths and weaknesses across various real-world scenarios.

Finally, the lack of universally accepted measures and definitions of synchrony leads researchers to use different computational and statistical approaches. This state of things reflects different aspects of interaction, which can introduce confusion [128].

For instance, the same definition might be given to distinct phenomena, and sometimes the same phenomenon is given multiple names [109]. This is the case for synchrony and coordination. Synchrony might be defined as specific in-phase coordination [83] whereas sometimes it represents both in-phase and anti-phase coordination [85].

## 2) ECOLOGICAL CONSIDERATION

Studies on human synchronization face notable challenges due to the inherent limitations of various data acquisition methods. Acquiring a signal that is meaningful regarding the phenomenon we are observing requires a trade-off between precision, ecological validity, and accessibility. For instance, force-sensitive pads and keyboards are cheap and provide high temporal precision for recording discrete movements, such as finger taps, but they reduce complex motor actions to single time points. This simplification overlooks the rich

dynamics of movement, such as position and speed variation, limiting a global understanding of motor behavior.

All acquisition techniques have pros and cons. For example, kinematic data collection is either restricted to specific body parts or needs lots of post-processing. As for the physiological signals, the delayed response of skin conductance adds complexity to the analysis of real-time synchronization. Considering neural data, EEG offers high temporal resolution but suffers from low spatial precision, while fMRI and fNIRS provide better spatial accuracy but are hindered by poor temporal resolution due to their reliance on hemodynamic responses.

Human synchronization abilities are regularly studied through finger-tapping tasks. This paradigm allows researchers to isolate how variations in stimuli affect different aspects of synchronization behavior, such as precision and stability. However, this setup may not fully reflect real-world conditions. For example, in these tasks, participants are asked to synchronize their movements with stimuli, whereas, in everyday life, individuals spontaneously synchronize with others. Additionally, in social interactions, individuals mutually and reciprocally adapt their behavior to each other, rather than fitting to a rigid temporal grid. Moreover, natural interactions are much more complex than simple laboratory tasks, which often focus on unidimensional motion and single stimulus modality. Finally, the nature of the stimuli in these experiments differs from what humans typically perceive and integrate in everyday life. For instance, sequences of artificially generated pure tones and flashing lights convey far less information than nonverbal cues typically exchanged during social interactions. Some studies suggest that humans are better at synchronizing with social stimuli than with non-social ones [67]. The authors' hypothesis for this difference is that individuals may be less engaged with metronomic pure tones or flashing lights than with more socially relevant stimuli. Alternatively, it could be that humans can perceive, extract, and use the richer information conveyed by nonverbal social behavior. While these explanations remain to be tested, the findings emphasize the importance of ecological validity in synchronization experiments.

Besides laboratory settings, measuring synchrony in natural interactions remains a challenge due to the multidimensional nature of the phenomenon. It encompasses the fine nonverbal cues, i.e., gestures, and facial expressions, but also neurophysiological responses, all of which interact and influence each other in complex ways. Additionally, individual differences such as personality traits, relationship history, familiarity, and the context of the interaction significantly impact synchrony patterns, making it difficult to generalize findings to the whole population [117].

Recent advances in the field of Artificial Intelligence open up new perspectives. First, expensive motion capture systems could be substituted by videos from webcams or smartphones, inertial sensors, and machine learning algorithms for pose estimation. This allows for lower cost set-up [127], and potentially better ecological validity for

field experiments [129]. Secondly, complex cues, such as speech and facial expressions could be taken into account by the multimodal approaches in the future. While advances in deep learning in different tasks show an integration of the various processing stages, we are wondering whether the next advances in the field of synchronization could be the integration of all steps (pre-processing, measures, interpretations) into one neural network model.

## B. DEVELOPING AUTOMATED METHODS

### 1) GLOBALLY MEASURING PERFORMANCE IN DIFFERENT CONTEXTS

In controlled settings, where participants are asked to perform rhythmic motions in a rhythmic context, relative phase measures provide a simple way to evaluate synchronization quality. This is because the target motion with which participants synchronize is well-defined. However, globally evaluating a performance in this scenario might already be difficult. The work by Bayd and al. [127] shows a panel of measures that output very different scores from the same performance on an SMS task. Ranging from local measurements to more global ones, and including the regularity index, the study stresses that a combination of these measures is necessary to capture the global quality of performance.

In [80], the authors compared different global measures of synchrony on performers practicing a Tai Chi routine side by side with human perception assessed by untrained participants. The performers were mocap recorded and two motion features have been tested to assess their synchrony. Participants were asked to score the movement synchrony between the two performers using a slider. They found that each tested measure correlated positively with human perception scores, however, authors argue that finding the measure that best relates to human perception might depend on the task analyzed. In more ecological, and by the way, more complex situations such as natural conversations or freestyle dancing, behaviors and motion are multimodal and unconstrained. As a result, assessing synchronization becomes even more challenging. Indeed, without an explicitly defined task, it is unclear what motion or features participants are synchronizing. Hence, objectively quantifying synchronization in such contexts requires some arbitrary decisions.

In that context, automatic synchrony analysis could be of great utility in self-finding the hidden and complex features humans may perceive and integrate when they interact. However, to evaluate the quality of similarity measures and to develop machine learning algorithms, a large quantity of well-annotated data is essential. However, among the gathered datasets, we found only 3 of them where the degree of synchrony, coordination, interaction, or entrainment is being annotated [24], [25], [95] and 3 where this synchronization is evaluated by untrained participants through either visual perception [80], [97], or auditory perception [96].

Multiple solutions can be proposed to address the lack of annotated datasets. The crowdsourcing approach has been developed in [130]. The authors developed an online questionnaire in which they uploaded various videos of human dancing. They then invited a large group of participants (432 individuals) to rate the perceived similarity and interaction between the dancers. This approach provided access to a substantial amount of annotated data with minimal labor required from each participant. Additionally, this data offers opportunities to deepen the understanding of synchronization perception and evaluate how different measures of synchrony relate to human perception.

Another complementary solution is to enhance collaborations across institutions. Sharing resources and access to diverse datasets can facilitate the adoption of a standardized annotation protocol and the creation of similar benchmark datasets. Furthermore, the use of machine learning (ML) to develop semi-automated annotation tools could greatly simplify the annotation process [131], [132]. These techniques offer a promising way to address the need for annotated datasets in this field.

Future contributions could focus on implementing supervised learning algorithms on such multimodal datasets to explain perceptual scores provided by both expert annotators and untrained participants.

### 2) ANNOTATING COMPLEX PHENOMENON

In complex and multimodal settings, for instance, when studying coordination inside a group of musicians, annotating might be a tedious task. In musical performance, musicians must show incredible SMS skills to play their instrument in time with the beat while also interacting socially with others, referring to multimodal and social cues to sync up their music together, especially in free improvisation styles. They might get insight from gazes, postures, and music to understand when it is their specific time to play. This context constitutes a unique blend of synchronization aspects and could benefit from expert annotation to assess the quality of the interaction, as defining objective measures remains challenging.

However, objectively annotating the quality of interaction and synchronization in this setup is not trivial. Indeed, the complexity of human behavior is hard to fully capture, going from moments of partial to full synchrony, with behaviors aligning on different time scales [115]. This means that annotators must deal with ambiguities, such as subtle or gradual changes in synchrony, leading to subjectivity in the labeling process and causing annotators to differ in their judgments. This can be mitigated by employing expert annotators and by establishing clear annotation guidelines as well as traditional inter and intra-agreement evaluation measures such as Cronbach's alpha [22] or Krippendorff's alpha [14].

### 3) EVALUATING A GLOBAL SCORE

Going beyond the challenge of reliably annotating such a complex phenomenon, developing automatic methods requires also good evaluation schemes. Moreover, distin-

guishing intentional synchrony from accidental overlap adds complexity, as true synchrony often implies mutual influence rather than mere coincidence [133]. The term “pseudosynchrony”, coined by Bernieri and Rosenthal [134], refers to coincidental behavioral events that may occur purely by chance during real interactions. To address the limitations of both human perception and automatic measures of detecting accidental overlap, a useful approach involves evaluating synchronization measures on surrogate data. Bernieri and Rosenthal [134] proposed synthesizing surrogate data by virtually pairing individuals who did not interact. These pseudo-interacting partners can serve as a baseline for evaluating synchrony measures, helping to distinguish intentional behavioral coordination from coincidental events arising from perception-action loops. Hudson et al. [125] replicated this methodology and used Cohen’s  $d$  to evaluate and assess the robustness of multivariate synchrony measures. In their approach, surrogate data for each sample is generated by swapping variables across multivariate time series, and the measure is recalculated on this modified dataset. The observed values are then compared to the surrogate baseline, with Cohen’s  $d$  and the percentage change reported to determine the sensitivity of the metrics to changes in the synthetic data parameters.

Finally, most studies collected in this paper focus their analysis on capturing synchronization. However, from another viewpoint, it has been found that more synchrony is not always beneficial [103]. Indeed Mayo and Gordon argue that moving in and out of synchrony could even be a socially adaptive behavior [135]. Going further, the disruption of synchronization can be considered a key factor in individual and collective creative processes [136]. This shows that musical and dance improvisation is fertile ground for the in-depth study of complex synchronization. It also questions the relevance of actual measures to account for the phenomenon of de-synchronization.

## VII. CONCLUSION

Building toward a more comprehensive understanding of human synchronization, this work centralizes datasets relevant to its study. It simplifies their access to adjacent fields such as affective computing, MIR, and DIR and reinforces the connection between them. In light of the FAIR principles, which emphasize that datasets should be findable, accessible, interoperable, and reusable, this paper also supports a standardized approach to organizing and sharing data. By adhering to these principles, researchers can ensure that datasets not only serve their immediate purpose but also support broader interdisciplinary collaborations, facilitate reproducibility, and maximize their long-term utility in advancing the field.

Richness in the multiplicity of synchronization tasks, contexts, signals, and data being studied highlights the multifaceted aspect of human synchronization. This emphasizes the need for multimodal datasets to capture the full extent of this phenomenon. However, such datasets introduce

significant technical challenges, particularly in achieving precise temporal alignment between different data streams. This review also highlights the interdisciplinary nature of human synchronization research. Different fields, ranging from computational, and behavioral neuroscience, to motor control and psychology are investigating this phenomenon. Each of them brings distinct perspectives, and sometimes, slight variation in their definitions of synchronization. This variability reflects the need for a more unified conceptual framework to integrate findings across disciplines.

Finally, this work highlights the existing challenges of globally scoring the synchrony in real-world scenarios such as social interactions or performance. The limited availability of datasets that provide annotations on the quality of coordination, whether from individuals involved in the interaction or external observers, slows down the development of automatic methods. However, this gap presents an exciting opportunity for future research. By developing more comprehensive and multimodal datasets, with clear and robust annotation methods, researchers can better capture the subtleties of synchronization in complex situations. Furthermore, while the study of this phenomenon is most meaningful in real-world scenarios, sensitive information about individuals should not become publicly available. Privacy of individuals and FAIR science are compatible, it requires clear anonymization protocols and transparent consent processes. Additionally, representation biases across different populations in datasets must not be overlooked. As the quality of rhythmic synchronization, including tapping behaviors, varies across cultures [137], future research should prioritize diversifying data collection to more accurately capture a broad spectrum of rhythmic behaviors across various cultural and demographic groups, thus avoiding the reinforcement of Western-centric perspectives.

Collaborative and interdisciplinary efforts are needed to standardize the study of human synchronization. This starts with adopting common definitions, using the same quantification of synchronization measurements, establishing clear annotation processes, and providing accessible datasets for comparing different analyses. Overall, having a common framework would enhance collaboration, and reproducibility and globally improve our understanding of human synchronization.

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